NIALL CASSIDY

DISCUSSION

A2 WJEC COMPUTER SCIENCE

NIALL CASSIDYNDIDATE NO: 0109
CENTER NO: 71741

Section 1: Discussion (5/100 marks)s

• Identify a substantial problem that provides sufficient scope for the candidate to access the full range of marks

My A-Level Unit 5 Computer Science coursework will be based on a local second-hand auto dealership called Monaghan Motors. It currently stores and processes all its data such as test drives, car details, employees, and customers on a paper-based system. There are many problems with this outdated approach.

- **Security** of customer and employee data may easily be compromised. A paper-based system can easily have files that can be misplaced or lost, and consequently, found by unauthorised personnel. This could be a breach of rules and regulations in the UK such as the Data Protection Act of 2018 which can result in massive fines for Mullingan Motors. Additionally, these paper documents would only be stored in a locked drawers and cupboards. If even a low-level employee found a key or the drawers unlocked, he could edit, destroy, or even forge documents to the storage of data as there are no levels of access. There would be no trace of these actions in a paper-based system and extremely hard to find a culprit responsible for tampering without use of CCTV.
- **Backing up** Mullingan Motor's data is also extremely time consuming and labour intensive. All the forms would need to be reprinted and filled in by a member of staff. Mistakes would likely and staff would need to careful that handwriting is legible. Even then, the backups would need to be stored in a durable safe, as the paper medium would be susceptible to fire damage which is a relatively probable outcome with Mulligan Motor's frequent dealings with Petrol. The safe would also be required to be waterproof as the area they are located is historically prone to heavy rain and flooding.
- **Data inconsistency, redundancy, and data integrity** will also plague Mulligan Motor's paper-based system. Since the chances of losing forms is relatively high compared to electronic databases, staff might rewrite data multiple times on different forms making increasing data redundancy and taking space up. The employees of Mulligan Motors may also be vulnerable to communication errors between them when filling out forms, leading to, for example, two different times for a test drive being stored on the system. This is called data inconsistency and will severely harm Mulligan Motors organisational capabilities with its current system. The overall integrity of data would be low in their traditional filing system additionally. Issues such as illegible handwriting and spilt drinks will make data on the forms inaccessible and useless to the car dealership. In my proposed database solution, these issues will not occur on the proposed electronic medium.

- **Searching and sorting** Mulligan Motors records is extremely inefficient on the current system. A member of staff would need to look at a record and place it in a certain cabinet, in a certain position amongst other records based off the alphabetical order of the last names. This is time consuming process and susceptible to human error, as the record may be placed in the wrong cabinet. If this would occur, the employees at Mulligan Motors would require searching every cabinet and every form to find the record. As the second-hand car dealership grows, this solution to human error will become increasingly unpractical.